PREDICTING AUTOMOBILE FUEL CONSUMPTION LEVELS USING RANDOM FOREST ALGORITHM

NGUYỄN THANH ĐĂNG [1*], HOÀNG MAI ANH BẢO [2], LÊ QUỐC BẢO [3]
[1,2,3] Khoa Công Nghệ Điện Tử, Trường Đại học Công nghiệp Hồ Chí Minh

ABSTRACT

The article focuses on predicting fuel consumption for automobiles using the Random Forest algorithm. Key parameters used for prediction include MPG (Miles Per Gallon), horsepower, curb weight, engine revolutions per minute (RPM), and acceleration. Data collected from SSANGYONG, HYUNDAI, TOYOTA automobile manufacturers is divided into two parts after processing: the training set and the testing set to ensure the model's objectivity. The predictive model employs the Random Forest algorithm, capable of handling relationships between input and output variables. The training process is conducted on the training set, where the model learns patterns from known data. The accuracy of the model is evaluated using the R-squared coefficient for both the training and testing sets. To verify the algorithm's effectiveness, the team implemented it on Jetson Nano hardware, assessing the study's accuracy against other algorithms and publishing the fuel consumption rates of various car manufacturers.

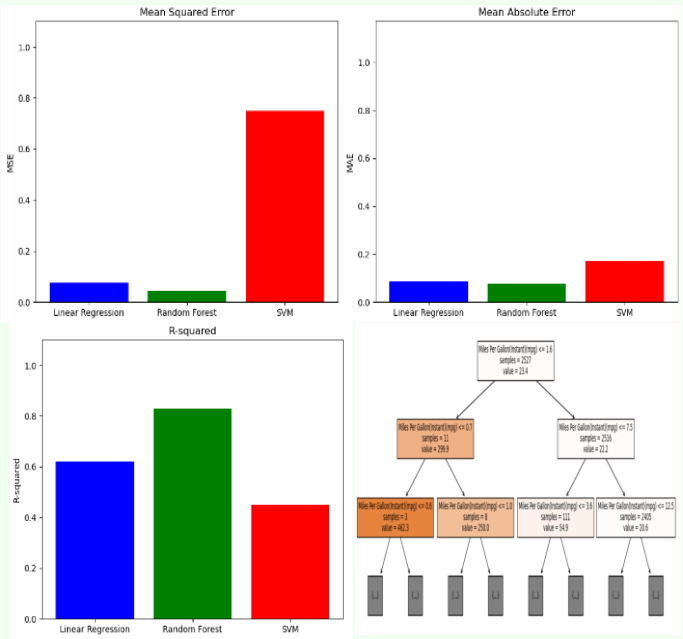

Figure 1: Error Comparison of Algorithms

Figure 2: Training Graph for Random Forest Algorithm

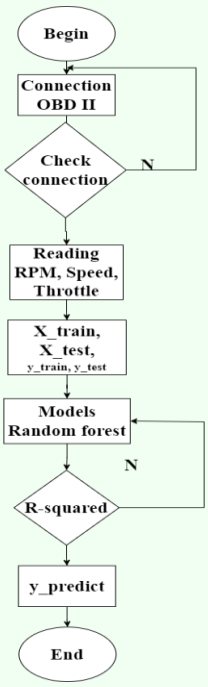

Figure 3: Data Collection, Training

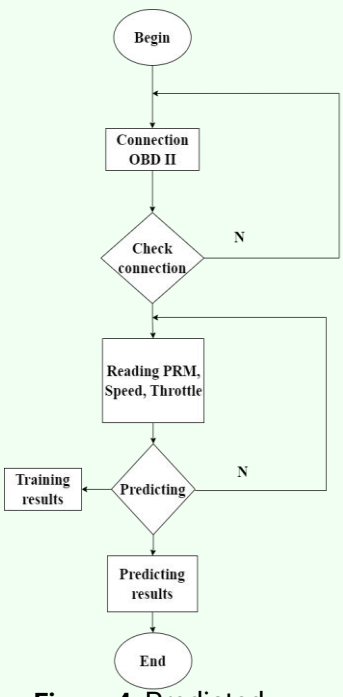

Figure 4: Predicted results

STT	Engine RPM	Speed(km/h)
1	751.5	0
2	751.5	0
3	751.5	0
4	749.75	0
5	750.25	0
6	750.25	0
7	1340.5	0
…	….	…
20438	1133	22
20439	1300.25	26
20440	1486	27
20441	1588	32
20442	1204.25	32
20443	1206.75	32
20444	1235	33
20445	1326	36
20446	1351.5	36
20447	1342.75	36

Figure 5: Collect data from OBDII save to file csv

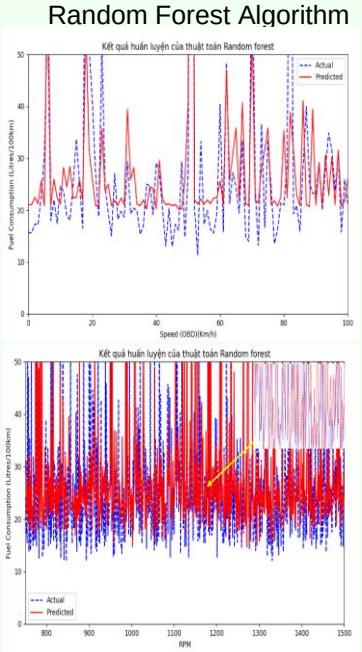

Figure 6: Random Forest training results

Introduce

Automotive fuel consumption prediction has become an important research area in recent times, with many studies focusing on using OBDII data to predict fuel consumption. Prediction methods have used various techniques, including neural networks, machine learning, and deep learning methods such as LSTM, with results showing high prediction potential and practical applications. However, many studies still lack detailed information on the model construction process and performance evaluation under real-world conditions. This study further develops a fuel consumption prediction method using random forest algorithms, improving performance and stability based on real-world driving data, aiming to optimize fuel management and reduce environmental impact.

Figure 7: Display of Predicted Fuel Consumption & Actual Fuel Consumption of Operating Vehicle